

10 R&D majors in India

1 Delta India Electronics

Delta India has state-of-the-art R&D centres located in Gurgaon and Bengaluru. Besides core product area and applications, it is working on complete solutions for smart-grid applications and automotive applications. In core product area and applications, it is constantly enhancing the efficiency rate of power converters and infrastructure monitoring solutions for telecom sites.

Delta offers highly customised products and R&D facilities in India work on indigenisation of UPS and display solution products suited to India and SAARC customer requirements and environment. Developing custom power supply for storage and network equipment is another core area for Delta India's R&D activity.

2 John F. Welch Technology Centre

GE's John F. Welch Technology Centre (JFWTC) in Bengaluru is a multi-disciplinary R&D centre accelerating the company's delivery of advanced technology to its global customers. The centre collaborates with GE's three other R&D facilities that form the GE Global Research team to conduct research, development and engineering activities for all of GE's diverse businesses worldwide.

Inaugurated on Sept 17, 2000, the centre is home to state-of-the-art laboratories working in the areas of mechanical engineering, electronic and electrical system technology, ceramics and metallurgy, catalysis and advanced chemistry, chemical engineering and process, polymer science and new synthetic materials, process modeling and simulation, power electronics and analysis technologies.

The centre has filed for more than 185 patents for R&D activities in Bengaluru and been granted twelve to date. In addition to GE's global research activities, JFWTC is also home to technology teams from other GE organisations including GE Advanced Materials, GE Consumer & Industrial, GE Energy, GE Transportation and GE Healthcare.

"The centre in Bengaluru does R&D for the entire spectrum of GE products. At this centre, we are working on CdTe technology for solar cells, which is being developed globally," informs Dr Mano Manoharan, general manager, operations, GE Global Research, and technology leader, manufacturing and materials, John F. Welch Technology Centre.

3 Freescale Semiconductor India

Freescale Semiconductor designs and manufactures embedded processing technologies for automotive, consumer, industrial and networking markets and its India design centres build system-on-chips for these markets along with creation of digital and mixed-signal IPs and an impressive portfolio of low-tier to high-tier processor core platforms.

Freescale Semiconductor has R&D centres in Noida, Bengaluru and Hyderabad, employing more than a thousand highly skilled professionals who play a significant part in delivering innovative products. Freescale India has been leading many important projects. It contributed immensely to the 360-degree surround-view parking assistance system launched recently, and also to small cell development for femto and pico base stations in the networking market.

4 Infineon India

Infineon India has a well established global R&D centre set up in Bengaluru. The company is a global semiconductor innovator in energy efficiency, mobility and security, employing around 280 professionals in India. Infineon's business spans from automotive electronics semiconductor solutions for power train and safety in two- and four-wheel vehicles, and chip card and security solutions for ePassport, citizen cards and drivers' licences to industrial and multi-market sector solutions for electronic ballasts, power supplies, and solar and wind energy conversion.

The R&D centre plays an essential role in software and hardware development for global products. Software development projects involve all layers from low-level software design to high-level application software and complex configuration software, whereas hardware design involves complex system-on-chip verification and validation and design automation systems (design flow and design libraries).

Infineon India plays a leading role in global automotive software strategy development such as automotive network standards, automotive open system architecture (AUTOSAR), automotive safety and electric vehicle motor control. Smart card operating systems and applications is another major area of R&D at Infineon India.

New chip designs are verified and validated to ensure full and perfect compliance to specification and delivered to the market within tight consumer windows. Parts of complex chip design flows and methodologies that reduce chip development time and ensure first-time right silicon are addressed.

5 Intel India

Intel India is Intel's largest non-manufacturing site outside of the US and considered a microcosm of Intel with the presence of major business groups. Having started its sales office in 1988 and R&D activities in 1999, Intel India has grown from less than 200 employees in 2000 to over 3000 employees now. A majority of these employees are in R&D. The Intel R&D centre in Bengaluru is spread across three facilities.

Intel India has made significant contributions through its engineering capability in silicon design, validation and systems software to a number of Intel Xeon server products and integrated